TPJC 2008 Prelim H2 Chem Paper 1: Multiple Choice Questions

1	C	**11**	D	**21**	A	**31**	B
2	B	**12**	D	**22**	B	**32**	A
3	C	**13**	A	**23**	A	**33**	B
4	B	**14**	C	**24**	A	**34**	D
5	C	**15**	B	**25**	B	**35**	A
6	D	**16**	C	**26**	A	**36**	C
7	A	**17**	A	**27**	D	**37**	A
8	C	**18**	C	**28**	D	**38**	B
9	D	**19**	A	**29**	A	**39**	B
10	B	**20**	A	**30**	A	**40**	A

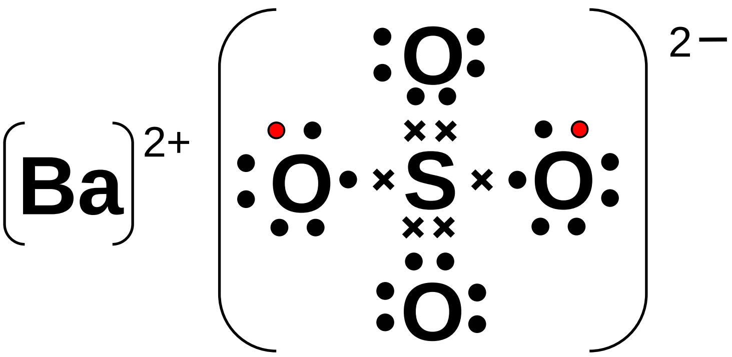
TPJC 2008 Prelim H2 Chem Paper 2: Structured

Question 1

a) i) White precipitate of $BaSO_4$ observed.
ii) $[SO_4^{2-}] = 0.240$ mol dm^{-3} $\qquad$ $[Ba^{2+}] = 0.479$ mol dm^{-3}
Ionic product = $[Ba^{2+}][SO_4^{2-}] = 0.479 \times 0.240 = 0.115$ mol^2 dm^{-6}
Since the ionic product > K_{sp}, a precipitate forms.

b) Charge density of Mg^{2+} is higher than Ca^{2+} due to a smaller size.
Mg^{2+} is more able to polarise the anion, resulting in $MgSO_4$ decomposing more easily.

c)



Question 2

a) at the cathode: $\mathbf{Al^{3+}(l) + 3e^- \rightarrow Al(l)}$ $\qquad$ at the anode: $\mathbf{O^{2-}(l) \rightarrow \frac{1}{2}O_2(g) + 2e^-}$

b) Moles of e^- = 48000/96500 = 0.497 mol
Mass of Al produced = 0.497/3 x 27.0 = **4.47 g**
Volume of oxygen gas liberated at s.t.p. = 0.497/4 x 22.4 = **2.78 dm^3**

c) i)

$\boxed{2}$ $Al^{3+}(g)$ + $\qquad$ $\boxed{3}$ $O^{2-}(g)$

$\boxed{2}$ $Al(g)$ $\qquad$ $\boxed{3}$ $O(g)$

$Al_{\boxed{2}}O_{\boxed{3}}(s)$

$\boxed{2}\,Al(s)$ $\boxed{\frac{3}{2}}\,O_2(g)$

ii) ΔH_2 : **sum of 1st, 2nd & 3rd ionisation energies of aluminium**

$= (577 + 1820 + 2740) = +5137\ kJ\ mol^{-1}$

ΔH_3 : **enthalpy change of atomisation of oxygen**

$= \dfrac{496}{2} = +248\ kJ\ mol^{-1}$

iii) ΔH_5 : **lattice energy of Al_2O_3**

$= +322 - [+644 + 3(+248)] - [2(+5137) + 1971)]$

$= -13\ 311\ kJ\ mol^{-1} = -1.33 \times 10^4\ kJ\ mol^{-1}$

<u>Question 3</u>

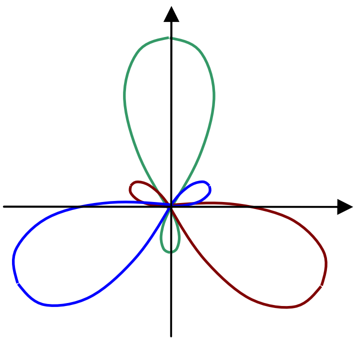
a) i) In compounds, the <u>d-orbitals</u> in the transition metal is <u>split into two energy levels</u> (or d-orbitals are not degenerate)
<u>Electrons from the d-orbital of lower energy can move to a vacant d-orbital of higher energy by absorbing energy</u> (d-d transition).
The <u>energy absorbed corresponds to the energy of visible light</u>.
The <u>colour observed is the complement of the colours absorbed</u>.

ii) Mn^{2+} / observe colour change from purple to colourless

b) $5C_2O_4^{2-} + 2MnO_4^{-} + 16H^{+} \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

c) Amount of $MnO_4^{-} = 0.0205 \times (4.88 \times 10^{-5}) = 1.00 \times 10^{-6}$ mol
Amount of $C_2O_4^{2-} = 5/2 \times (1.00 \times 10^{-6}) = 2.5 \times 10^{-6}$ mol
Amount of Ca^{2+} in 1 cm^3 of blood = <u>**2.5×10^{-6} mol**</u> [$Ca^{2+} \equiv C_2O_4^{2-}$]
Mass of Ca^{2+} in 1 cm^3 of blood = $(2.5 \times 10^{-6}) \times 40.0 = 1.00 \times 10^{-4}$ g = 0.100 mg
Concentration of Ca^{2+} = 0.100 x 100 = <u>**10.0 mg per 100 *ml***</u>

d) i) sp^2 hybridisation



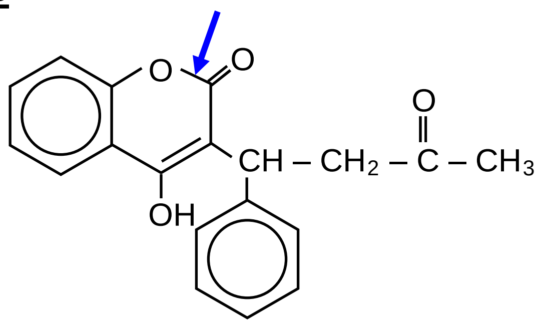
ii) 120°

Question 4

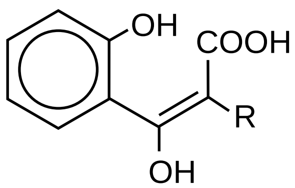
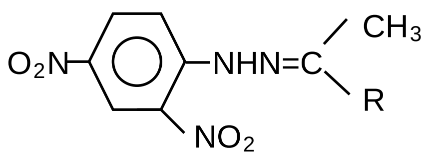
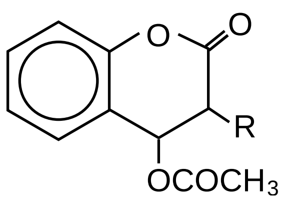
Reactant	Observations	Type of reaction	Equations
(a) hot aqueous NaOH	pale yellow chlorine solution decolourised	disproportionation	$6NaOH + 3Cl_2 \rightarrow$ $5NaCl + 3H_2O + NaClO_3$
(b) aqueous KI	brown solution formed	displacement	$Cl_2 + 2KI \rightarrow I_2 + 2KCl$
(c) propane, in the presence of light	white fumes of HCl observed	free radical substitution	$C_3H_8 + Cl_2 \rightarrow C_3H_7Cl + HCl$
(d) phenylamine	white precipitate	electrophilic substitution	$C_6H_5NH_2 + 3Cl_2 \rightarrow C_6H_2NH_2Cl_3 + 3HCl$

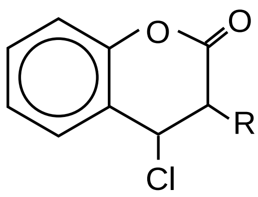
Question 5

a)

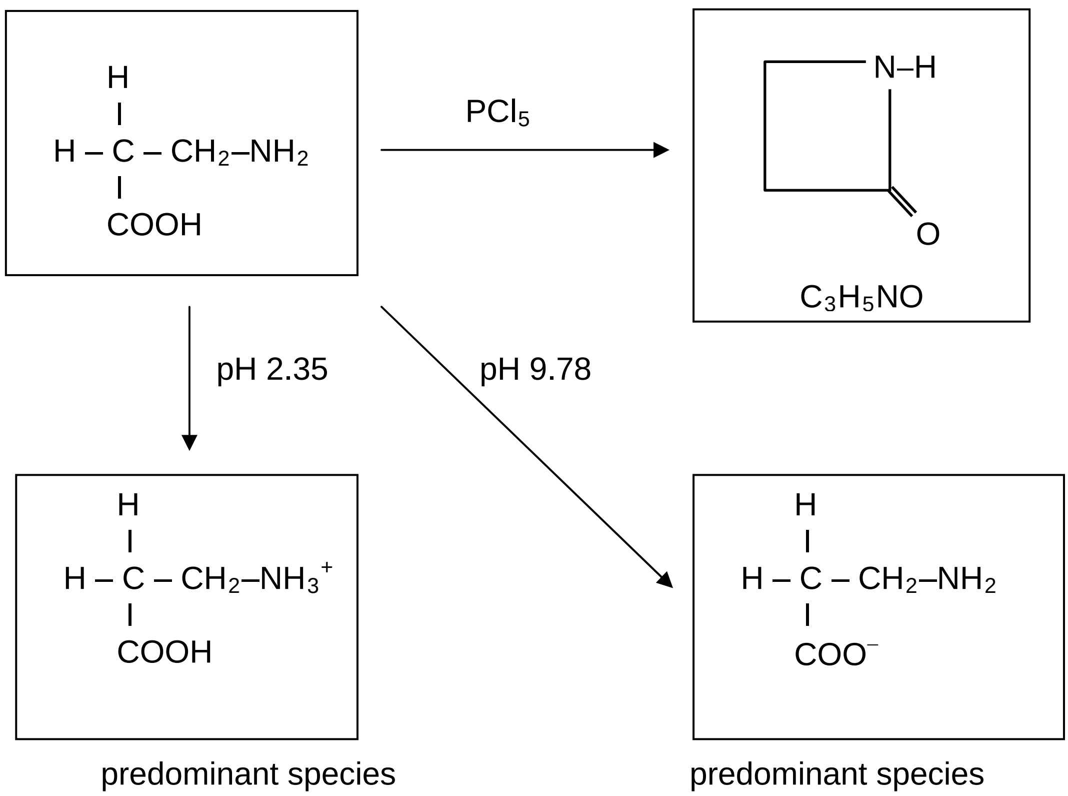


b)

reagent	group responsible	structural formula(e) of the organic product(s)
alkaline I_2(aq)	$-\overset{\overset{O}{\|}}{C}-CH_3$ [1] (methyl carbonyl)	$R-\overset{\overset{O}{\|}}{C}-O^- + CHI_3$ [½ + ½]
H_2SO_4(aq) under reflux	$-\overset{\overset{O}{\|}}{C}-O-$ [1] (ester)	 [2]
2,4 – DNPH	$R-\overset{\overset{O}{\|}}{C}-R$ [1] (carbonyl)	 [1]
ethanoyl chloride	–OH [1] (alcohol)	 [1]

phosphorous pentachloride	–OH (alcohol) [1]		[1]

(c)



TPJC 2008 Prelim H2 Chem Paper 3: Free Response

Question 1

a) i) Oxidising Agent – substance which is able to accept electrons

Equation : $\mathbf{Cl_2 + SO_3^{2-} + H_2O \rightarrow 2Cl^- + SO_4^{2-} + 2H^+}$

In this reaction, the oxidation state of Cl has decreased from 0 to –1, showing that it has gained electrons.

(OR the oxidation state of S has increased from +4 to +6, showing that it has lost electrons).

ii) Add a few drops of $AgNO_3$(aq).

Formation of a white precipitate would indicate the presence of the Cl^-(aq) formed.

b) From the Data Booklet,

$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$ $\quad E^\ominus = +1.36$ V

For the reaction between Cl_2 and $N_2H_5^+$, $E^\ominus_{cell} = 1.36 + 0.17 = +1.53$ V

Since $E^\ominus_{cell} > 0$ V, the reaction is expected to be feasible.

Equation : $\mathbf{N_2H_5^+ + 2Cl_2 \rightarrow 4Cl^- + N_2 + 5H^+}$

c) i) electrophile : Br^+ free radical : Br

ii) Mechanism : **free radical substitution**

Initiation

$Br\text{–}Br \xrightarrow{u\nu} 2Br\bullet$

(or state that homolytic fission occurs in presence of uν)

Propagation

$CH_3COOH + Br\bullet \rightarrow \bullet CH_2COOH + HBr$

$\bullet CH_2COOH + Br_2 \rightarrow BrCH_2COOH + Br\bullet$

Termination

$Br\bullet + Br\bullet \rightarrow Br_2$

$Br\bullet + \bullet CH_2COOH \rightarrow BrCH_2COOH$

$\bullet CH_2COOH + \bullet CH_2COOH \rightarrow HOOCCH_2\text{–}CH_2COOH$

("combination of free radicals")

d) i) HF forms hydrogen bonding between its molecules, resulting in the H^+ being less likely to be dissociated.

ii) $K_a = \dfrac{[H^+][F^-]}{[HF]}$

$[H^+]^2 = 0.200 \times 5.62 \times 10^{-4}$

pH = **1.97**

iii) Moles of HF after addition of NaOH = (0.1 x 0.200) – (0.05 x 0.100)
= 0.015 mol

[HF] after mixing = 0.015/0.15 = 0.1 mol dm^{-3}

Moles of NaF present after addition of NaOH = 0.005 mol

$[F^-] = 0.005/0.15 = 0.0333$ mol dm^{-3}

$[H^+] = (5.62 \times 10^{-4})(0.1) / (0.0333) = 0.00169$ mol dm^{-3}

pH = **2.77**

iv) $\mathbf{F^-(aq) + H_2O(l) \rightleftharpoons HF(aq) + OH^-(aq)}$

K_b of $F^- = K_w/K_{a\ of\ HF} = 10^{-14} / (5.62 \times 10^{-4}) = 1.78 \times 10^{-11}$ mol dm^{-3}

$[OH^-]^2 = (1.78 \times 10^{-11}) \times 0.0022$

pOH = 6.70 ⇨ pH = 14 – 6.7 = **7.3**

Question 2

a) Rate constant is the proportionality constant in the experimentally-determined rate equation.
Half-life refers to the time taken for the reactant concentration to fall to half of its original value.

b) i) **Rate = *k*[RBr][NaOH]**

Units of *k* = $\mathbf{mol^{-1}\ dm^3\ time^{-1}}$ [*time in any units*]

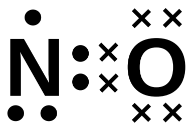
ii) When [RBr] x 2, rate of reaction x 2.
Since the [RBr] and reaction rate increased by the same extent, half-life of RBr will remain the same at **40 min**.

When [NaOH] x 4, rate of reaction x 4.
Since [RBr] remained constant but reaction rate increased 4 times, RBr will be used up at a faster rate. Thus, half-life of RBr will be 40/4 = **10 min**.

iii) $RBr + NaOH \xrightarrow{\text{slow}} ROH + NaBr$

c) i) Due to the higher temperature, the reaction proceeded at a faster rate.
ii) Dynamic equilibrium has been achieved.
iii) At higher temperature T_2 , the % of NO in the mixture at equilibrium is lower.
This implies that the equilibrium shifted right at higher temperature.
Since high temperatures favour endothermic reactions, the forward reaction must have been endothermic.
Therefore ΔH is positive.
iv) For curve III, the reaction rate was faster but the same % of NO was present at equilibrium (or same equilibrium position but achieved more quickly).
This could have been achieved with the addition of a catalyst.

d) i)



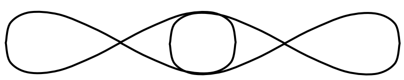
ii) dative bond / NO possesses a lone pair of electrons
iii) oxidation state of Fe = +3 $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

Question 3

a) i) $H_2S + 3/2O_2 \rightarrow SO_2 + H_2O$
ii) Vol. of H_2S = 0.005/100 x 1000 = 0.05 dm^3 ; Vol. of SO_2 produced = 0.05 dm^3
iii) $CaCO_3(s) + SO_2(g) \rightarrow CaSO_3(s) + CO_2(g)$
$CaSO_3(s) + \frac{1}{2}O_2(g) \rightarrow CaSO_4(s)$

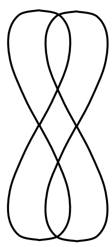
b) In H_2S, there are 2 bond pairs and 2 lone pairs of electrons around S.
This results in a bent shape with bond angle about 104^o.
In SO_2 , there are 2 bond pairs and 1 lone pair of electrons around S.
This results in a bent shape with bond angle slightly < 120^o.

c) **σ-bond** is a covalent bond formed when orbitals overlap head-on.



σ-bond

π-bond is a covalent bond formed when orbitals overlap sideways.



π-bond

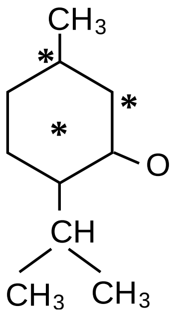
d) CO_2 has a simple molecular structure consisting of molecules held together by induced dipole-dipole attraction.
SO_2 has a simple molecular structure consisting of molecules held together by permanent dipole-dipole attraction.
Since induced dipole-dipole attraction are weaker than the permanent dipole-dipole attraction, less energy is required to separate the CO_2 molecules.

e) i) Activation energy is the minimum energy reactant molecules must possess in order to have effective collision.
A catalyst lowers the activation energy of a reaction by providing an alternative reaction pathway of lower activation energy.
This enables the catalyst to increase the reaction rate.

ii) The activation energy of the reaction is high due to the requirement for the collision of two negatively-charged ions.
Iron(III) ions catalyse the reaction as such:
$2Fe^{3+} + 2I^{-} \rightarrow 2Fe^{2+} + I_2$
$2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$
Iron(III) ions are recovered at the end of the reaction.

Question 4

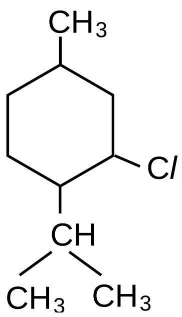
a) i)



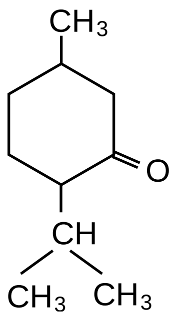
Each chiral carbon gives rise to 2 optical isomers, therefore giving a total of 2^3 = **8** stereoisomers

ii) The molecule exists as enantiomers and rotates plane-polarised light.

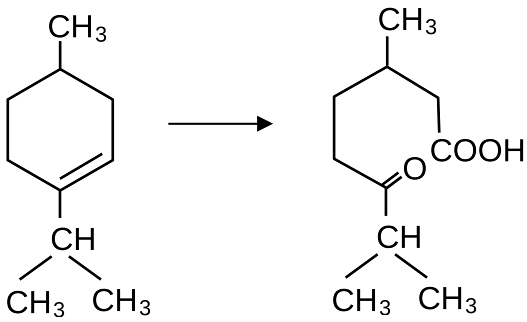
b) i)



S



U



T **V**

[*Note that formation of the less substituted alkene as **T** will yield a corresponding structure for **V** which DOES NOT agree with the given molecular formula. As such, the answer will NOT be accepted.*]

ii) elimination or dehydration
iii) alcoholic KOH; reflux
iv) Add 2,4-dinitrophenylhydrazine at room temperature.
Formation of orange-yellow precipitate shows presence of the carbonyl.

c) Concentration of menthol = (40/1000)/0.320 = **0.125 g dm^{-3}**

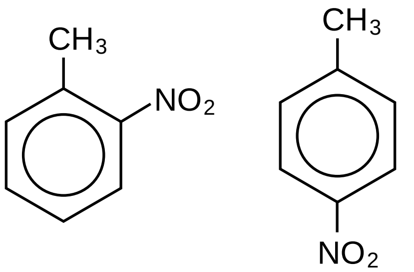
d) i) Menthol has a simple molecular structure consisting of molecules held together by hydrogen bonds.
n-Undecane has a simple molecular structure consisting of molecules held together by induced dipole-dipole attraction (or van der Waals' forces).
Since induced dipole-dipole attraction are weaker than hydrogen bonds, less energy is required to separate the n-Undecane molecules.
ii) Both menthol and ethanol are able to form hydrogen bonds with water molecules.
However, the larger hydrophobic alkyl group of menthol reduces its ability to interact with the water molecules, leading to reduced solubility.

e) Energy required to boil the water = 500 x 4.18 x 75 = 156 750 J
Energy to be provided by combustion of menthol = 100/75 x 156 750 = 209 000 J
= 209 kJ

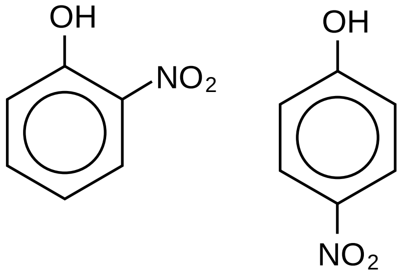
Moles of menthol required = 209/6310 = 0.0331 mol
Mass of menthol required = 0.0331 x 156 = **5.16 g**

Question 5

a) i)

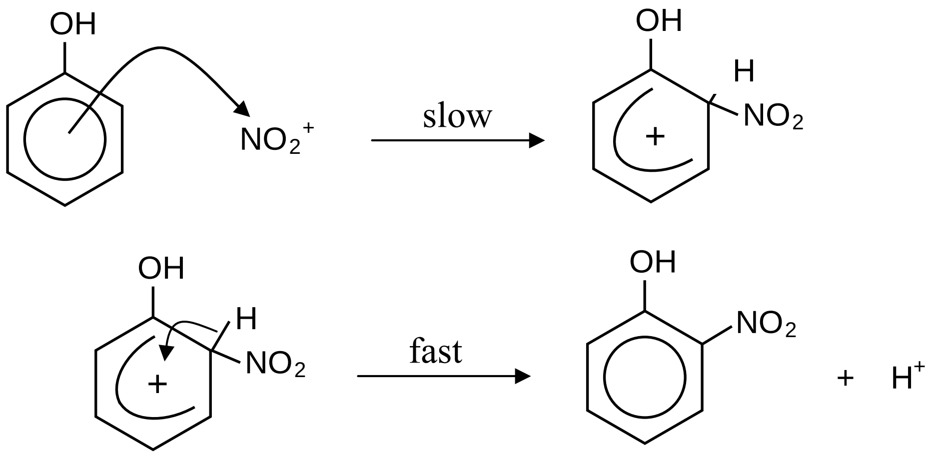


P & Q



R & S

ii) electrophilic substitution



iii) The benzene ring in phenol is more reactive towards nitration.
This is due to the –OH group being electron-donating and causing the benzene ring to be more electron-rich and susceptible to electrophilic attack.

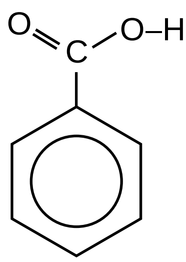
iv) 4-nitrophenol is the major product due to stearic considerations.

v) The nitrated phenol is expected to be a stronger acid.
The electron-withdrawing $-NO_2$ group disperses the negative charge on the nitrated phenoxide ion.
Thus the nitrated phenoxide ion is more stable than the phenoxide ion.
This results in the nitrated phenol donating the H^+ more readily.

b) Both catechol and hydroquinone form intermolecular hydrogen bonds.
However, catechol is also able to form intramolecular hydrogen bonds.
Formation of intramolecular hydrogen bonds in catechol reduces the extent of intermolecular hydrogen bonding.
This leads to weaker intermolecular forces in catechol and therefore less energy is required to overcome these forces.

c) i) Reflux with acidified $KMnO_4$(aq).
Methylbenzene : decolourisation of $KMnO_4$
Ethylbenzene : decolourisation of $KMnO_4$ with bubbles of gas (CO_2)

Organic Product formed in both cases:



ii) Boil with NaOH(aq). Acidify with HNO_3, followed by $AgNO_3$(aq).
C_6H_5Br: no precipitate formed
$C_6H_5CH_2Br$: cream precipitate (AgBr) formed

Organic Product formed: